

Conditionals & If-Else Statements

 RAMRODZ ROBOTICS · TEAM 6366 · PROGRAMMING WORKSHEET



Name _____ Date _____

  RAM RODZ ROBOTICS

TEAM #6366

HILLCREST & WOODMONT HIGH SCHOOLS · SIMPSONVILLE, SC

 **FOCUS CHECK-IN**

 Struggling

 Okay

 Ready

 **TODAY'S BIG QUESTION**

How does a robot decide WHAT to do based on what's happening around it?

P PARKING LOT — RANDOM THOUGHTS

If your brain jumps to a question or idea, write it here so you don't lose it — ask later!

1 Key Words — Check Off When Explained

☐ **Condition**

A yes/no question: "Is the button pressed?"

☐ **if statement**

"IF this is true → do this"

☐ **else if**

"OR IF this other thing is true..."

☐ **Boolean**

A value that's only **true** or **false**

☐ **else statement**

"OTHERWISE → do this instead"

☐ **Comparison Operator**

Symbols that compare two values:

`== != < >`

2 The if Statement

- ☒ Listen: instructor explains
if

JAVA — FRC EXAMPLE

```
if (buttonPressed) {  
    shooterSpeed = 0.8; // fire shooter  
}
```

 **Plain English:** IF the A button is pressed → run the shooter at 80%

- ☒ I can see where the condition is (inside the parentheses)
Hint: it's what's between the ()

- ☒ I can see where the action is (inside the curly braces { })

TRY IT — FILL IN THE BLANK

Complete this sentence: IF _____, THEN run the intake motor.

Adding

3 else

— The "Otherwise"

- ✓ Listen: instructor explains
else

JAVA — FRC EXAMPLE

```
if (buttonPressed) {  
    shooterSpeed = 0.8;    // button pressed → fire!  
} else {  
    shooterSpeed = 0.0;    // button NOT pressed → stop  
}
```

💡 Plain English: **IF** A button → shoot | **OTHERWISE** → stop

- ✓ The robot always does *one* of two things — never both, never nothing
- ✓ else
runs only when the
if
condition is **false**

4

Multiple Choices with

else if

- ✓ Listen: instructor explains
else if

JAVA — DRIVE SPEED CONTROL

```
if (throttle > 0.8) {  
    driveSpeed = 1.0;    // full power  
} else if (throttle > 0.4) {  
    driveSpeed = 0.5;    // half power  
} else {  
    driveSpeed = 0.0;    // stop  
}
```

💡 The computer checks top-to-bottom and **stops at the first true condition**

- ✓ Only ONE branch runs, even if multiple conditions are true
- ✓ Order matters! More specific conditions go first

QUICK QUESTION

If
`throttle = 0.9`
 , which branch runs above? Write it here:

5 Comparison Operators Cheat Sheet

- ✓ Listen: instructor reviews comparison symbols

Symbol	Means	Robot Example	Result
<code>==</code>	equals	<code>teamNumber == 254</code>	true / false
<code>!=</code>	not equals	<code>mode != "auto"</code>	true / false
<code>></code>	greater than	<code>rpm > 500</code>	true / false
<code><</code>	less than	<code>temp < 90</code>	true / false
<code>>=</code>	greater or equal	<code>score >= 10</code>	true / false
<code><=</code>	less or equal	<code>timeLeft <= 30</code>	true / false

WRITE YOUR OWN

Write a condition using any operator above. (It can be silly — what matters is the syntax!)

Robot Scenario — Put It Together

SCENARIO

Our robot needs logic for the shooter:

- If the **Shoot button** is pressed AND we're within range → fire the shooter
- Else if only **in range** (no button) → spin up the flywheel to prep
- Otherwise → everything off

SKETCH IT — FLOWCHART OR CODE (YOUR CHOICE)

☒ I attempted the scenario (even a partial answer counts!)

☒ **END OF LESSON — I CAN DO THIS**

☒ Explain what a condition is in my own words

☒ Read an
`if`
statement and say what it does

☒ Explain the difference between
`else`
`and`
`else if`

☒ Name at least 3 comparison operators

☒ Write a simple
`if-else`
for a robot action